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Research Article

Aeroelastic Instability Problems for Wind Turbines

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mechanical vibration

This paper deals with the aeroelastic instabilities that have occurred and may still occur for modern commercial wind turbines: stall-induced vibrations for stall-turbines, and classical flutter for pitch-regulated turbines. A review of previous works is combined with derivations of analytical stability limits for typical blade sections that show the fundamental mechanisms of these instabilities. The risk of stall-induced vibrations is mainly related to blade airfoil characteristics, effective direction of blade vibrations and structural damping; whereas the blade tip speed, torsional blade stiffness and chordwise position of the center of gravity along the blades are the main parameters for flutter. These instability characteristics are exemplified by aeroelastic stability analyses of different wind turbines. The review of each aeroelastic instability ends with a list of current research issues that represent unsolved aeroelastic instability problems for wind turbines.

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Introduction

The state-of-the-art in aeroelastic stability of wind turbines is introduced in this paper. The aim is to give the reader a fundamental understanding of the most important instabilities due to the interaction of the elastic vibrations of wind turbines and the unsteady aerodynamic forces acting on them. This fundamental understanding will enable the reader to pursue deeper insight of aeroelastic stability of wind turbines based on the relevant literature reviewed in this paper.

The two aeroelastic instabilities expected to occur, or have been observed for commercial wind turbines, can be divided into two categories: *stall-induced vibrations* and *classical flutter*, where the first category of instabilities is better known as *stall flutter* in aeronautics and turbomachinery. These two categories of instabilities are linked to the two wind turbine concepts that dominate the market: active-stall-regulated turbines and pitch-regulated, variable-speed turbines.

Stall-regulated turbines operate with their blades in a separated flow and have experienced stall-induced vibrations.[1,2] In the second section of this paper, it is shown how the special airfoil characteristics in stall can lead to negative aerodynamic damping of blades vibrating with certain directions relative to the rotor plane.[3–12] The direction of blade vibration depends not only on the blade stiffness distributions and pre-twist, but also on the turbine dynamics due to interactions of flap- and edgewise *rotor whirling modes*.[13,14] Tower bending modes of stall-regulated turbines may also have low, or negative damping because the blades vibrate unfavorably relative to the rotor plane. Generally, blade vibrations close to the rotor plane, as in edgewise whirling modes and lateral tower modes, are often least damped; but flapwise and longitudinal tower modes can also

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be lightly damped if the blade airfoils have abrupt stall characteristics. These issues are exemplified by two case studies of actual stall-induced vibration problems for stall-regulated turbines in the end of the second section of this paper.

Pitch-regulated turbines do not operate in stall, thereby the risk of stall-induced vibrations is minor; except at standstill and around rated wind speeds for turbines with high-performance rotors operating close to stall. The aerodynamic damping of flapwise rotor and longitudinal tower modes is high, whereas the damping of edgewise whirling, drivetrain torsion, and lateral tower modes is low due to the low aerodynamic coupling. Long slender blades of pitch-regulated turbines operating in attached flow may have the risk of classical flutter if the frequency ratio between flapwise bending and torsional modes is sufficiently low, the rotor speed is sufficiently high, and the center of mass is sufficiently aft on the blade cross sections.[15–17] The instability mechanism of flutter is described in the third section of this paper, and a new case study of flutter limits for the 5 MW pitch-regulated reference turbine proposed by NREL[18] shows that it may be sufficient to consider a single isolated blade when these flutter limits are determined, as also found in a previous study.[17]

Besides the two aeroelastic instabilities stall-induced vibrations and flutter, modern commercial turbines include one, or several controllers that may lead to instabilities. For example, the constant power controller at above-rated wind speeds leads to negative damping from the generator to the drivetrain, whereby pitch-regulated turbines often have active drivetrain damping built into the generator controller.[19] Controller-induced instabilities lie outside the scope of this paper.

Wind Turbine Terminology and Modal Dynamics

Understanding of the aeroelastic instabilities considered in the following sections requires a common terminology for the dynamics of wind turbines. An introduction to a terminology often used for the degrees of freedom and mode shapes of three-bladed turbines is therefore given based on a repetition of the modal dynamics of three-bladed turbines previously introduced by the author.[13]

Degrees of Freedom

Figure 1 illustrates the terminology used for the degrees of freedom of wind turbines. The tower may bend in the *longitudinal* (fore-aft) and *lateral* (sideward) directions (parallel and perpendicular to the wind direction), whereby the nacelle and rotor are translated relatively and rotated in *tilt* and *roll*, respectively. Torsion of the tower will result in *yaw* rotation of the nacelle and rotor. Rotation of the drivetrain results in *azimuth* rotation of the rotor. In this paper, the azimuth angle of a blade is zero in the downward position of the blade. On active-stall and pitch-regulated turbines, the blades of the rotor can be rotated in *pitch* about the pitch axis extending spanwise perpendicular to the blade root flange. The blade pitch is often defined as zero when the chord of the airfoil close to the tip lies in the rotor plane. Bending of the blades is defined in *flapwise* (flatwise) and *edgewise* (lead-lag) directions, which at zero pitch correspond to out of and in rotor plane directions. Finally, the blades may also deform in *torsion* (twist), whereby the airfoil cross sections are rotated. Elastic deformation of the airfoils during bending and torsion is often neglected in aeroelastic analyses.

Modal Dynamics

Figure 2 shows a typical example of the lower-order modal dynamics of a three-bladed turbine described by the natural frequencies of the first 10 modes of a 600 kW turbine with an asynchronous generator plotted as function of rotor speed in a Campbell diagram.[13] The behavior of rotor speed normalized frequencies and corresponding mode shapes is similar to that of a 5 MW turbine, except that the drivetrain torsional modes on modern variable-speed turbines have different boundary conditions and therefore different natural frequencies.

To understand the modal dynamics of a three-bladed turbine, it is convenient to consider the modal dynamics of its most flexible components: tower, drivetrain and blades.

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lots of
movement types

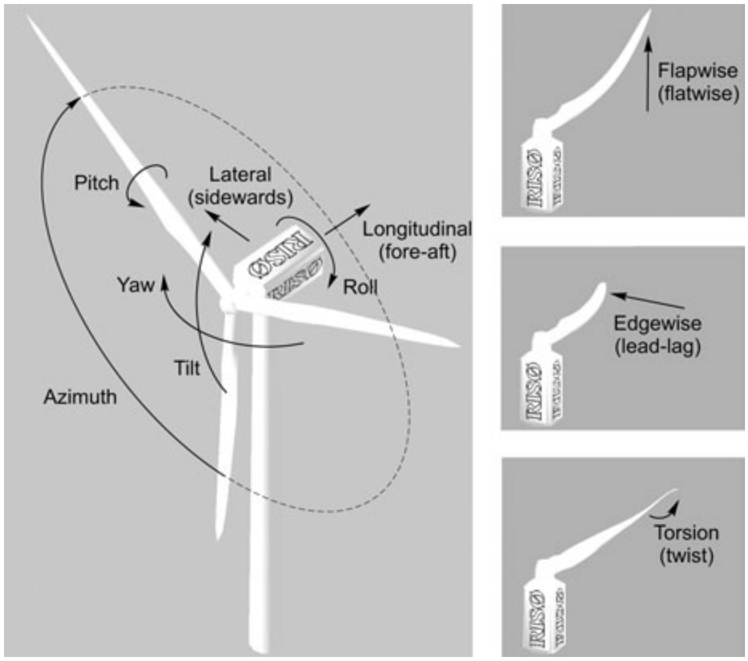

Figure 1. Terminology for degrees of freedom of wind turbines

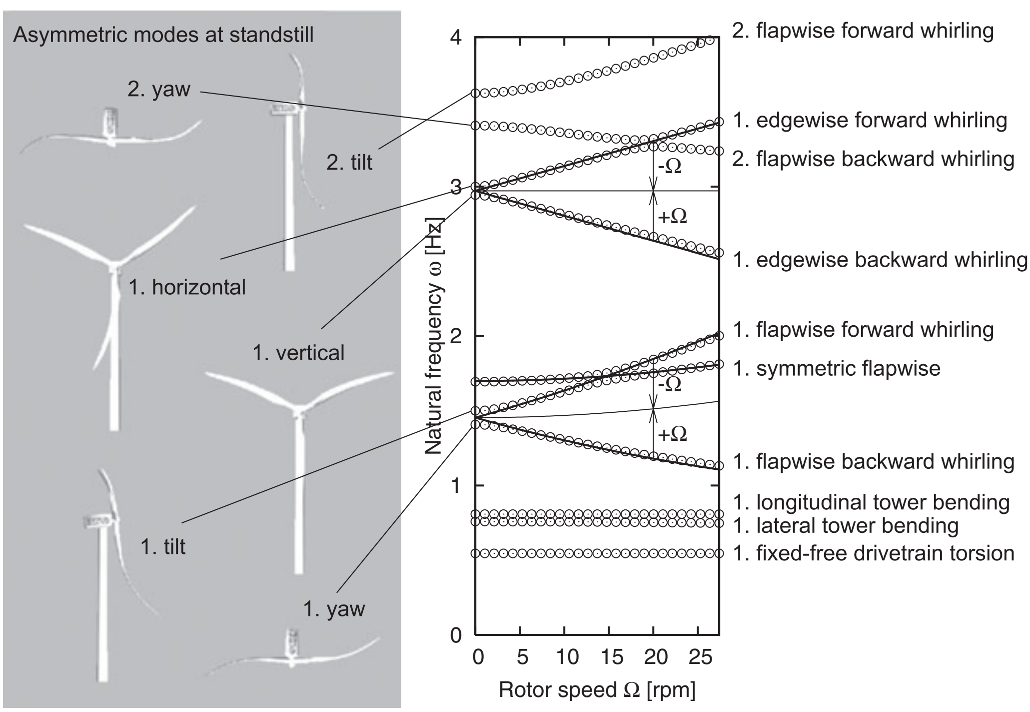

Figure 2. Campbell diagram of ground-fixed natural frequencies (○) for the first 10 structural modes of a 600 kW three-bladed turbine with illustrations of asymmetric standstill modes.[14] Lines denote approximations to the computed frequencies and the center frequencies of the rotor whirling modes given ±Ω added to the ground-fixed natural frequencies

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The first pair of tower lateral and longitudinal bending modes are represented in the lower-order turbine modes (modes 2 and 3 in Figure 2, *numbered by standstill frequencies*); the second pair of tower bending modes may also be represented for high towers used on low wind sites. The flexibility of the nacelle and rotor has little influence on the tower bending modes. The first torsional mode of the tower may be represented in the lower-order modes for turbines with strutted towers because of their low torsional stiffness. The natural frequencies of the structural tower modes are practically independent of the rotational speed of the rotor; however, there is a rotor speed dependent amplitude ratio and phase between the lateral and longitudinal components of their mode shapes due to the gyroscopic coupling of tower top motion and the rotor.

The drivetrain consisting of main shaft, gearbox and generator introduces a mode in the lower-order turbine modes corresponding to torsion between the rotor hub and generator (mode 1 in Figure 2). This drivetrain torsional mode is coupled with simultaneous edgewise bending of all blades, and it is affected by the generator control. Constant speed control and asynchronous generators are modeled by a fixed torsional boundary condition, whereby the drivetrain torsional mode is coupled with lateral tower bending. Constant torque and power controls are modeled as free torsional boundary conditions, whereby the drivetrain modal frequencies are increased. The torsional drivetrain mode is not represented for turbines with direct-drive multi-pole generators; instead, there is a mode consisting of symmetric edgewise bending of the three blades.

Each blade contributes to the modal turbine dynamics with their bending and torsional modes (see examples of blade modes in Figure 1). Vibrations of the three rotor blades in a combination of their mode shapes are coupled to the vibrations of the remaining turbine through the reaction forces at the rotor hub. To analyze this coupling, it is necessary to describe the motion of the rotor blades and thereby the reaction forces in the ground-fixed inertia frame where the motion of the hub is defined.

The *Coleman transformation* is a method to describe the motion of the individual blades in the ground-fixed frame (see e.g. Johnson[20] and Peters[21]). The fundamental assumption of this method is that the rotor is *isotropic*, i.e. the three blades are identical and symmetrically mounted on the hub. Let q_1, q_2 and q_3 be the blade coordinates for an identical *deflection shape* of each blade, then the vibration of each blade in this deflection shape is assumed to be given by

$$q_k = a_0 + a_1 \cos\left[\Omega t + \frac{2\pi}{3}(k-1)\right] + b_1 \sin\left[\Omega t + \frac{2\pi}{3}(k-1)\right] \quad k = 1, 2, 3 \tag{1}$$

where k is the blade number, Ω is the mean rotor speed, and a_0, a_1 and b_1 are three *multi-blade coordinates* replacing the blade coordinates q_k. Note that $\Omega t + \frac{2\pi}{3}(k-1)$ is the mean azimuth angle to blade number k at time t. If the blade coordinates q_k describe a flapwise blade deflection shape, then a_0 describes a simultaneous flapwise motion of all three blades, while a_1 and b_1 describe tilt and yaw motions, respectively. If the coordinates q_k describe an edgewise blade deflection shape, then a_1 and b_1 describe horizontal and vertical motions, respectively, and a_0 describes a simultaneous edgewise blade motion that may interact with the drivetrain torsional mode. Figure 3 shows the asymmetric rotor motions given by a_1 and b_1 for flapwise and edgewise blade deflection shapes with arrows indicating the reaction forces on the remaining turbine.

Transformation (1) of the rotor motion and reaction forces into the ground-fixed frame enables the formulation of an eigenvalue problem for three-bladed turbines with isotropic rotors.[13] An eigensolution of such eigenvalue problem represents a mode of the turbine, where the natural frequency ω is given by the eigenvalue and the mode shape is given by the eigenvector in multi-blade coordinates $a_0 = A_0 \sin(\omega t + \phi_0)$, $a_1 = A_a \sin(\omega t + \phi_a)$ and $b_1 = A_b \sin(\omega t + \phi_b)$. Substitution of this mode shape into Equation (1) shows that the mode shape in blade coordinates becomes

$$q_k = A_0 \sin(\omega t + \phi_0) + A_{\mathrm{BW}} \sin\left[(\omega + \Omega)t + \frac{2\pi}{3}(k-1) + \phi_{\mathrm{BW}}\right] + A_{\mathrm{FW}} \sin\left[(\omega - \Omega)t - \frac{2\pi}{3}(k-1) + \phi_{\mathrm{FW}}\right] \tag{2}$$

where the new modal amplitudes A_{BW} and A_{FW}, and phases ϕ_{BW} and ϕ_{FW} are given by the modal amplitudes A_a and A_b, and phases ϕ_a and ϕ_b of the cyclic multi-blade coordinates in the eigenvector.

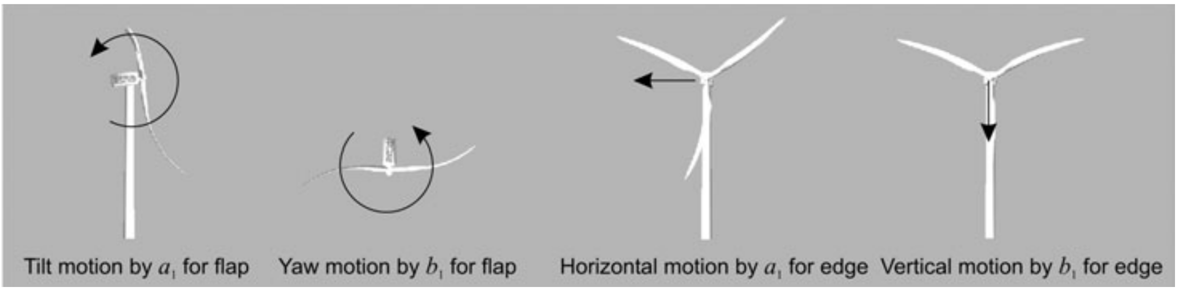

Figure 3. Asymmetric rotor motions given by a_1 and b_1 for flapwise and edgewise blade deflection shapes with arrows indicating the reaction forces on the remaining turbine

Equation (2) shows that the modal response of blade k may consist of three components: a symmetric component where all blades deflect simultaneously in the considered deflection shape with the amplitude A_0, and two cyclic components where the blades deflect with phase shifts of $\frac{2\pi}{3}$ and the amplitudes A_{BW} and A_{FW}, respectively. As indicated by the subscripts, the cyclic components represent *backward whirling* (BW) and *forward whirling* (FW) of the rotor blades, where the blades vibrate in a backward and forward sequence relative to their numbering; the whirl direction is determined by the sign of the phase shifts $\frac{2\pi}{3}(k-1)$ for each blade $k = 1, 2, 3$. The frequency of the symmetric modal component is ω, while the frequencies of the whirling modal components are shifted by $\pm\Omega$ in the rotating blade frame. The reason for these frequency shifts is that the natural frequency ω is given in the ground-fixed frame; an observer on the tower top (in the ground-fixed frame) will measure the natural frequency ω for any mode. The observer on a blade will also measure the frequency ω_0 if the mode is purely symmetric ($A_0 \neq 0$ and $A_{BW} = A_{FW} = 0$), but for a pure BW mode ($A_{BW} \neq 0$ and $A_0 = A_{FW} = 0$) the blade observer will measure the frequency $\omega_{BW} + \Omega$, and for a pure FW mode ($A_{FW} \neq 0$ and $A_0 = A_{BW} = 0$) the blade observer will measure the frequency $\omega_{FW} - \Omega$ in the rotating blade frame. A turbine mode is rarely pure, but has a combination of symmetric, BW and FW components, whereby the blade observer may associate up to three frequencies with this particular mode.

Returning to the modal dynamics of the 600 kW turbine in Figure 2, the modes 4, 5, 6, 9 and 10 (still numbered by standstill frequencies) are related to the first flapwise blade bending mode, whereas modes 7 and 8 are related to the first edgewise bending mode of the blades.

At standstill, the edgewise mode pair (7 and 8) are the first vertical and horizontal turbine modes as shown by the illustrations in Figure 2. Herein, the reaction forces from the vertical and horizontal rotor motion are balanced by in-phase vertical and horizontal motions of the rotor hub. Their natural frequencies are slightly different due to the different vertical and horizontal flexibilities of the remaining turbine supporting the rotor, which also causes these natural frequencies to be slightly below the natural frequency of the first edgewise blade bending mode of 3 Hz (the frequencies would asymptotically coincide with this edgewise blade frequency as the vertical and horizontal stiffnesses of the rotor support approach infinity).

When the rotor is rotating, the edgewise mode pair couples in the first edgewise BW and FW modes. Their natural frequencies split almost perfectly in the ground-fixed frame by $\pm\Omega$ around a center frequency close to the edgewise blade frequency of 3 Hz, as shown by the lines in Figure 2. This splitting can be explained by transforming these ground-fixed frequencies ω_{BW} and ω_{FW} into the rotating blade frame using equation (2). Assuming that A_{BW} and A_{FW} are the dominating components of the respective mode shapes, the frequencies of the BW and FW modes in the rotating blade frame are $\omega_{BW} + \Omega$ and $\omega_{FW} - \Omega$, respectively. Hence, the blade observer measures the same frequency $\omega_{BW} + \Omega = \omega_{FW} - \Omega$ for the two whirling modes, where this center frequency is the natural frequency of the edgewise blade mode that forms these two turbine modes.

Modes 4 and 5 represent a similar coupling of asymmetric standstill modes into whirling modes with similar splitting of their natural frequencies in the ground-fixed frame. Here, the first flapwise bending mode of the blades couples in the first yaw and tilt modes at standstill, and in the first flapwise BW and FW modes when the rotor is rotating. Their natural frequencies at standstill are distinct and lower than the natural frequency of the flapwise blade mode (1·64 Hz) due to the distinct yaw and tilt flexibility of the rotor support. Their ground-fixed frequencies split by $\pm\Omega$ around a center frequency which now increases with rotor speed due to *centrifugal stiffening*.*

The centrifugal stiffening effect on the flapwise blade frequency is extracted from the natural frequency of the first symmetric flapwise mode (mode 6). This symmetric turbine mode consists of simultaneous deflection of all three blades in the first flapwise blade mode (A_0-component), whereby the natural frequencies in the ground-fixed and rotating blade frames are equal. A small longitudinal tower motion in counter-phase balances the reaction force at the hub, and the natural frequency at standstill (1·69 Hz) is slightly higher than natural frequency of the flapwise blade mode due to the reduced vibrational inertia of the blades. Because the centrifugal stiffness is proportional to the square of the rotor speed, a fit of the expression $\sqrt{\omega_f + s\Omega^2}$ to the natural frequency of the first symmetric flapwise mode yields the centrifugal stiffness coefficient s and the standstill frequency ω_f (cf. Figure 2).

Note the small deviation of the natural frequencies of the symmetric and FW flapwise modes from the approximating frequency lines in Figure 2 as the two modes cross over around 15 rpm. Here, the two modes couple in mode shapes, i.e. they both contain A_0- and A_{FW}-components for the flapwise deflection of the rotor blades.

The whirling coupling of asymmetric standstill modes with $\pm\Omega$ frequency splits is not as dominant for the second yaw and tilt modes (modes 9 and 10) as for the first yaw–tilt and vertical–horizontal mode pairs (see Figure 2). The most important reason is that their natural frequencies at standstill are more separated due to different couplings to the flexibility of the remaining turbine. Both first and second yaw–tilt mode pairs are related to the first flapwise blade mode (the second flapwise mode has a natural frequency of 4·8 Hz), but for the first yaw–tilt modes the remaining turbine vibrates in phase with the rotor motions, whereas the remaining turbine vibrates in counter-phase with the rotor motion for the second yaw–tilt modes (leading to lower effective inertia and therefore higher frequencies). This counter-phase also means that the vibrations of the rotor blades stay closer to the rotor plane, whereby the gyroscopic reaction forces at the rotor hub are smaller and contribute less to the frequency splitting.

Note the crossing of the first edgewise FW mode and second flapwise BW mode about 20 rpm. The turbine is operating at 27 rpm, but there is still a coupling of these two edgewise and flapwise whirling modes during operation. The edgewise BW mode obtains a flapwise component that changes its aerodynamic damping, as shown in a following case study of stall-induced vibrations.

Stall-Induced Vibrations

Stall-induced vibrations of wind turbine blades became an issue during the continuous optimization and upscaling of stall-regulated wind turbines. The first example of stall-induced vibrations was measured in 1980 on the stall-regulated *Nibe* A turbine with rated power about 600 kW.[1] Further examples of stall-induced vibrations[2,22] lead to extensive investigations which have provided a detailed understanding of the problem.[3–13]

Figure 4 shows a measured edgewise blade root bending moment on a 500 kW stall-regulated turbine, where a large edgewise blade vibration rises above the gravity loads after about 300 s.[6] The frequency of this vibration corresponds to the natural frequency of the first edgewise blade bending mode which indicates an aeroelastic instability.

*The centrifugal stiffening effect is not significant for the edgewise blade bending mode where the centrifugal forces also have a softening effect on the linear stiffness of deflection in the rotor plane that almost cancels the stiffening effect. The softening effect is a geometric artifact of the linear component of the centrifugal forces in the edgewise direction for this blade deflection in the rotor plane and perpendicular to the pitch axis.

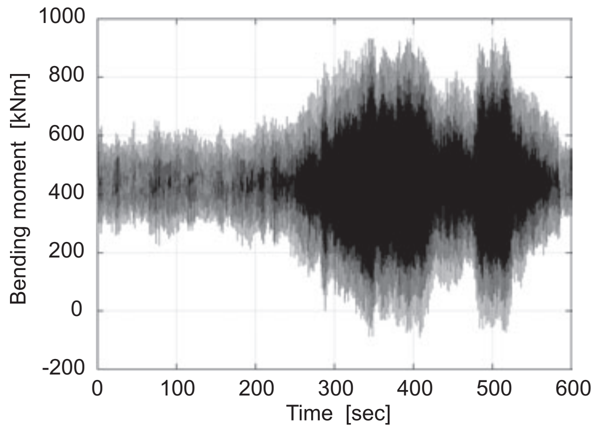

Figure 4. Stall-induced edgewise blade vibrations measured in edgewise bending moment at the blade root of a 500 kW turbine at 23·2 m s^{-1} mean wind speed. Taken from Petersen et al.[6]

The purpose of this section is to introduce the most important aspects of stall-induced vibrations that have been identified in the previous studies. The three parameters that dominate the risk of stall-induced vibrations are:

1. *Airfoil characteristics*. If the blades have airfoils with abrupt stall characteristics, the risk of stall-induced vibrations is larger for the turbine.
2. *Direction of vibration*. Dependent of the airfoil characteristics, there are directions of blade vibrations relative to the rotor plane where the risk of stall-induced vibrations is larger. Directions of blade vibrations depend on the entire turbine dynamics.
3. *Structural damping*. If a turbine mode has a slightly negative aerodynamic damping due to stall-induced vibrations, it may be compensated by structural damping.

These fundamental statements are explained in the following. First, a typical section model with quasi-steady aerodynamics is used to show the effects of the airfoil characteristics and the direction of vibration relative to the rotor plane. Then, two case studies are presented where the risk of stall-induced vibrations has been an issue. These examples introduce the concept of an *effective direction of vibration* to handle the fact that the blades on a wind turbine are interacting with each other and the remaining turbine,[13] whereby the blades vibrate in combinations of their mode shapes. Finally, the section ends with a list of the current issues on stall-induced vibrations.

Instability Mechanism for a Typical Section

To understand the instability mechanism of stall-induced vibrations, it is sufficient to consider a typical blade section with one degree of freedom, subjected to quasi-steady aerodynamic forces as illustrated in Figure 5. The translation of the airfoil $x(t)$ is defined by the direction of vibration θ relative to the rotor plane. The quasi-steady aerodynamic forces L and D per unit-length can be determined for the steady-state inflow angle ϕ_0 and relative speed W_0, and the steady-state angle of attack α_0 as follows:

$$L = \frac{1}{2}\rho c W^2 C_L(\alpha) \quad \text{and} \quad D = \frac{1}{2}\rho c W^2 C_D(\alpha) \tag{3}$$

where c is the chord length, ρ is the air density, W is the relative speed, α is the angle of attack, and C_L and C_D are the lift and drag coefficients evaluated at α. The relative speed and the angle of attack are functions of the velocity of the blade section:

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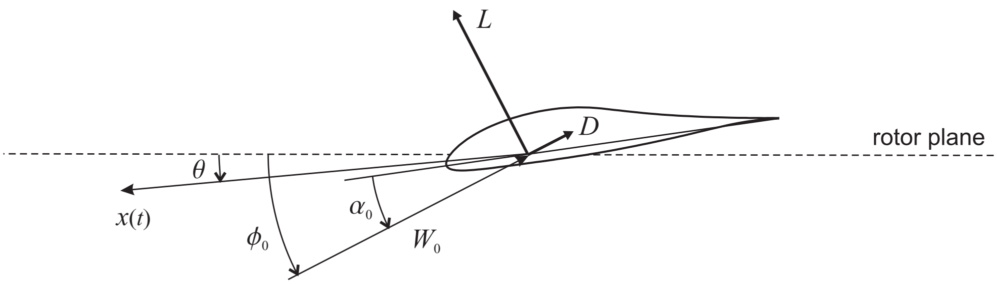

Figure 5. A typical blade section with one degree of freedom x(t) in the direction of vibration θ relative to the rotor plane and subjected to quasi-steady aerodynamic forces L and D determined by the steady-state inflow angle ϕ_0, relative inflow speed W_0, and angle of attack α_0

$$W = \sqrt{(W_0 \cos\phi_0 + \dot{x}\cos\theta)^2 + (W_0 \sin\phi_0 + \dot{x}\sin\theta)^2} \\ \alpha = \phi - \phi_0 + \alpha_0 \tag{4}$$

where $\dot{x}$ is the time derivative of x, and ϕ is the inflow angle relative to the rotor plane:

$$\phi = \arctan\left(\frac{W_0 \sin\phi_0 + \dot{x}\sin\theta}{W_0 \cos\phi_0 + \dot{x}\cos\theta}\right). \tag{5}$$

Projection of the aerodynamic forces onto the direction of vibration leads to the force:

$$F_x = \frac{1}{2}\rho c W^2 C_L(\alpha)\cos(\phi - \theta) - \frac{1}{2}\rho c W^2 C_D(\alpha)\sin(\phi - \theta). \tag{6}$$

By substituting the flow variables in Equations (4) and (5) into this expression, it is seen that the aerodynamic force is a function of the velocity $\dot{x}$ as the only independent variable.

Linearization of the aerodynamic force (Equation (6)) using Taylor expansion about $\dot{x} = 0$ leads to the approximation $F_x \approx F_0 - \eta\dot{x}$, where the coefficient is given by

$$\eta = \frac{1}{2}c\rho W_0[C_D(3 + \cos(2\theta - 2\phi_0)) + C_L'(1 - \cos(2\theta - 2\phi_0)) + (C_L + C_D')\sin(2\theta - 2\phi_0)] \tag{7}$$

where the lift and drag coefficients and their gradients $C_L' = dC_L/d\alpha$ and $C_D' = dC_D/d\alpha$ are evaluated at $\alpha = \alpha_0$. The coefficient in Equation (7) corresponds to the damping coefficient of a viscous damping term approximating the aerodynamic forces in the equation of motion for $x(t)$. If this aerodynamic damping coefficient is negative, the aerodynamic forces add energy to vibrations of the blade section, whereby its steady-state position becomes unstable if the amount of structural damping is insufficient to dissipate this energy.

Several fundamental statements can be deduced from the aerodynamic damping coefficient (equation (7)) which characterize the instability mechanism behind stall-induced vibrations:

1. The aerodynamic damping is proportional to the relative speed of the steady-state inflow W_0, and not to the square of the relative speed as the aerodynamic forces.
2. The first term in the brackets shows that drag always increases the aerodynamic damping because the coefficient to C_D is always positive $3 + \cos(2\theta - 2\phi_0) \geq 2$. This effect is largest for vibrations parallel to the inflow ($\theta - \phi_0 = 0$), because it is related to variations in relative velocity.
3. The second term in the brackets shows that a negative lift gradient decreases the aerodynamic damping because the coefficient to C_L' is positive, or zero $1 - \cos(2\theta - 2\phi_0) \geq 0$. This destabilizing effect is largest for vibrations perpendicular to the inflow ($\theta - \phi_0 = \pm\frac{\pi}{2}$) because it is related to variations in angle of attack.

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4. The third term in the brackets shows that the positive lift and drag gradient of airfoils in normal operation decreases the aerodynamic damping for directions of vibrations in the second and fourth quadrants relative to the inflow ($\frac{\pi}{2} < \theta - \phi_0 < \pi$ and $-\frac{\pi}{2} < \theta - \phi_0 < 0$).

Figure 6 illustrates the important ranges of directions of vibration relative to the inflow direction given by the three terms of the aerodynamic damping coefficient (Equation (7)). The arrows in each range indicate 'pull', or 'push' of the most critical direction of vibration with lowest damping due to each of these three terms: drag $C_D > 0$ will push the critical direction away from the inflow direction, a negative lift gradient $C_L' < 0$ will pull it perpendicular to the inflow, and the sum $C_L + C_D' > 0$ will pull towards a 45° angle to the inflow. Given a state-steady inflow angle ϕ_0 and an angle of attack α_0 determining these airfoil characteristics, the sign of the aerodynamic damping can be derived from all three terms of Equation (7) as function of the direction of vibration θ.

Figure 7 shows two examples of the aerodynamic damping coefficient (Equation (7)) for two steady-state angles of attack on the NACA 63-415 airfoil. A local speed ratio of 6 is chosen, whereby $\phi_0 \approx 9{\cdot}5°$ if induced velocities are neglected. It corresponds to the 75% radial cross section of a blade operating at a tip speed ratio of 8. The 75% radial section is chosen because the aerodynamic forces are largest outboard on the blade. The

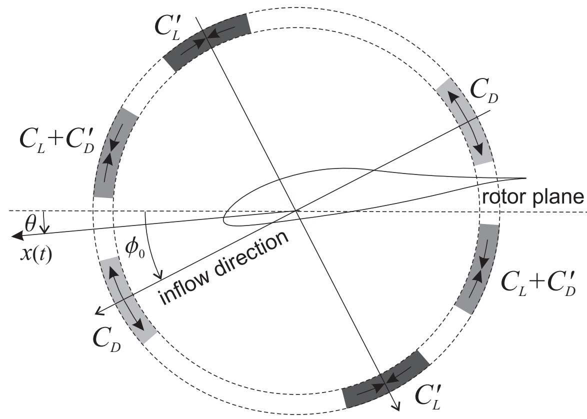

Figure 6. Important ranges of directions of vibration relative to the inflow direction given by the aerodynamic damping coefficient (Equation (7))

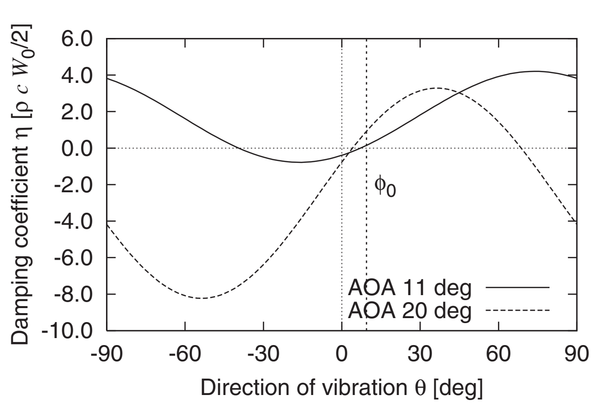

Figure 7. Aerodynamic damping coefficients (equation (7)) as function of direction of vibration for two angles of attack: 11° and 20°. Local speed ratio 6 ($\phi_0 \approx 9{\cdot}5°$) and NACA 63-415 airfoil

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NACA 63-415 airfoil is chosen to present a typical airfoil used outboard on blades of last generation turbines with low aerodynamic damping properties.

The 75% radial section of a pitch-regulated blade may operate around $\alpha_0 = 11°$ close to maximum lift for near-rated wind speeds. Figure 7 shows that the aerodynamic damping in this case is lowest for vibrations close to the rotor plane with a slight downwind motion in the forward airfoil motion. This direction is related to the term of Equation (7) dominated by the sum $C_L + C'_D$ (cf. Figure 6), because both C_D and C'_L terms are close to zero.

The 75% radial section of a stall-regulated blade may operate around $\alpha_0 = 20°$ at high winds. Figure 7 shows that the aerodynamic damping in this case is lowest for vibrations in an angle of −50° relative to the rotor plane, which is related to the terms of Equation (7) dominated by the sum $C_L + C'_D$ and the lift gradient C'_L (cf. Figure 6) due to the stall characteristics of the NACA 63-415. The aerodynamic damping coefficient is more negative than in the pre-stall case and therefore the name 'stall-induced vibrations'. Stall-regulated turbines with the NACA 63-415 airfoil may have problems with stall-induced vibrations; both turbines used in the following two case studies have NACA 63-415 airfoils on the outboard part of their blades.

The lift and drag coefficients for the clean NACA 63-415 airfoil measured by Bak *et al.*[23] are shown in Figure 8 with the label 'smooth'. The stall is fairly abrupt: maximum lift of 1·3 decreases to 0·6 in deep stall combined with a large drag gradient in the medium stall region. The stall can be smoothened by mounting stall-strips on the leading edge of the airfoil. Figure 8 shows measured lift and drag coefficients for triangular stall-strips (7 mm high on an airfoil with 600 mm chord) in different positions of the leading edge. By smoothening the stall, the aerodynamic damping coefficient can be increased due to the decreased maximum lift, lower lift and drag gradients, and the increased drag. Stall-strips have therefore been used in many cases as a practical solution to problems with stall-induced vibrations,[24] however, with the cost of reduced power production. To compensate for the reduced power production, stall-strips are often combined with mounting of vortex generators on the inboard part of the blades, whereby the lift and therefore aerodynamic torque of the rotor can be increased. The levels of vibration and aerodynamic forces are relative smaller at the inboard part than at the outboard parts of the blades, hence the inboard part with increased lift will not contribute significantly to the negative damping of the turbine modes.[24]

Dynamic Stall Effects

The above description of stall-induced vibrations using the quasi-steady aerodynamic damping coefficient (Equation (7)) is simplified to ease the understanding of the instability mechanism. There are other aspects of this instability that must be included when the risk of stall-induced vibrations is assessed for a wind turbine.

An important aspect of stall-induced vibrations was identified early by Rasmussen *et al.*[3] in one of the first studies of the instability, namely that *the use of quasi-static aerodynamics in wind turbine aeroelastic codes was shown to be deficient when calculating stalled rotor dynamic performance*. The unsteady aerodynamic

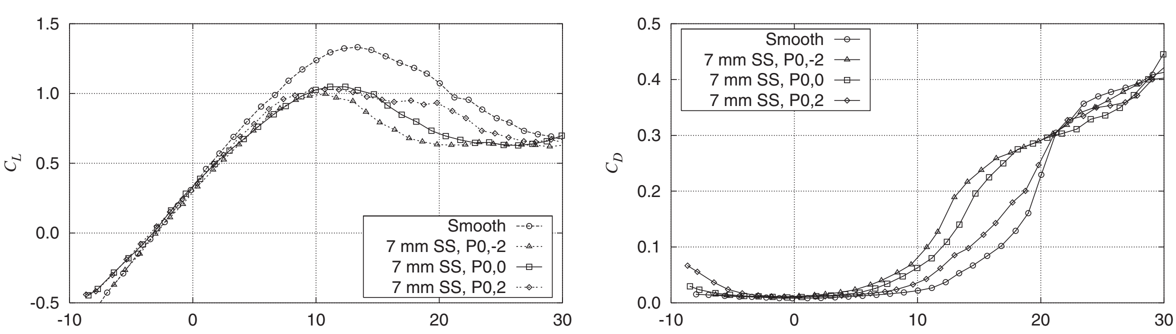

Figure 8. Measured lift and drag coefficients of the NACA 63-415 airfoil with stall-strips positioned at the stagnation point at −2, 0 and 2° angle of attack[23]

forces do not follow the quasi-steady assumption that lift and drag on the blade airfoils at any time are given by the instantaneous angle of attack through their lift and drag curves (polars) measured under static conditions. If the flow over an airfoil is stalled but not fully separated (not in deep stall) then a *dynamic stall* effect will cause the lift to momentarily increase after a step change in angle of attack, whereafter it will decrease to the static value at the new angle of attack. This dynamic stall effect therefore means that the dynamic lift of an oscillating airfoil will loop around the static lift curve as shown in Figure 9. Rasmussen *et al*.[4] have shown that these dynamic stall loops correspond to an increased effective lift gradient, whereby the aerodynamic damping of airfoil vibrations in a critical direction of vibration is increased compared to quasi-steady predictions.

Two Case Studies

This section contains case studies of two different stall-regulated turbines with the NACA 63-415 airfoil on the outboard part of their blades, where stall-induced vibrations are suppressed by the use of stall-strips. The first turbine is a MW-sized active-stall-regulated turbine, where the lateral tower bending mode is very low damped even with stall-strips mounted on the blades. This low damped mode leads to amplification of the lateral bending moment at the tower bottom; however, they are still smaller than the design loads from the longitudinal tower bending moment. The second turbine is the 600 kW stall-regulated turbine considered for the introduction to modal dynamics.[13] The blades of this turbine would experience stall-induced edgewise vibrations similar to the vibrations shown in Figure 4, if they were not fitted with stall-strips.

Stall-Induced Tower Vibrations

Although the stall-induced tower vibrations of the MW-sized active-stall-regulated turbine were not dominating the design loads of this turbine, the following case study is interesting because it shows many elements of this instability.

A stability analysis was performed for the particular turbine because the time simulations of the design load cases with the nonlinear aeroelastic wind turbine code HAWC[25] showed a tendency of lateral tower vibrations. Figure 10 shows the mean and standard deviations (SDs) of the lateral tower bottom bending moment from these simulations at different wind speeds. The mean values follow the expected trend: they increase until the rated wind speed about 12 m s^{-1}, whereafter they remain almost constant for increased wind speed. The SD should increase monotonously over the whole wind speed range because the excitation energy of the turbulence (for the same intensity) increases monotonously with the wind speed. However, this presumption requires that the dynamic amplification of this stochastic excitation is independent of wind speed, i.e. the damping of the tower vibrations remains constant over the whole wind speed range. The peaks of the SDs at 19 and 23 m s^{-1} therefore represent cases where the lateral tower damping is reduced.

To investigate the cause of these peaks in SDs, the power spectral density functions of the lateral tower bottom bending moment from the load case simulations are plotted in a waterfall plot in Figure 11. As expected,

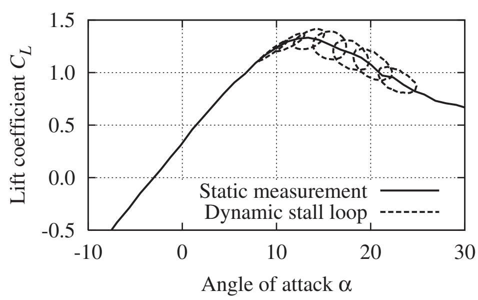

Figure 9. Dynamic stall loops for the NACA 63-415 airfoil measured by Bak et al.[23]

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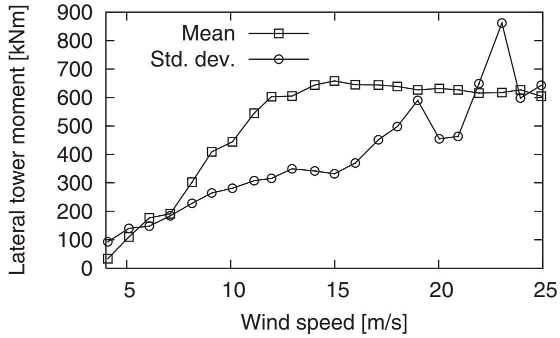

Figure 10. Mean and SDs of the lateral tower bottom bending moment from 10 min load case simulations at different wind speeds with HAWC[25]

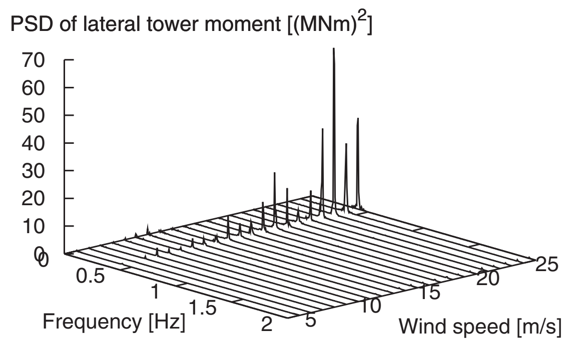

Figure 11. Power spectral density (PSD) functions of the lateral tower bottom bending moment from 10 min load case simulations at different wind speeds with HAWC[25]

there is a dynamic amplification of the stochastic excitation at the natural frequencies of the first tower bending modes around 0·45 Hz that increases with wind speed. However, there are significant peaks in the frequency response at the tower frequency around the same wind speeds as the peaks of the SDs of the lateral tower bottom bending moment. These significant amplifications can only be explained by a low, or even negative aeroelastic damping of the first tower bending modes. Analysis of the SDs of the longitudinal tower bending moment (not plotted) shows no significant peaks suggesting that the longitudinal tower bending mode is lightly damped which is confirmed by the stability analysis.

The eigenvalue analysis is performed with the linear aeroelastic stability tool HAWCStab[11] which uses the same input as the input to the nonlinear HAWC model of a wind turbine. This tool is used to set up and solve eigenvalue problems for any operational condition by linearizing the aeroelastic HAWC model about a steady-state equilibrium, where the steady aerodynamic forces due to a presumed uniform wind (no wind shear) without turbulence are balanced by a symmetric deflection of the rotor blades. The linear model includes the effects of dynamic stall using a modified Beddoes–Leishman model,[26] and the periodic terms in the equations of motion due to the rotor are eliminated using the Coleman transformation (cf. equation (1)).

Figure 12 shows the aeroelastic damping of the first two tower bending modes obtained from the real part of eigenvalues for the investigated turbine. The longitudinal mode has positive damping with a step in damping due to the step in the operational rotor speed at low wind speeds. The lateral mode has low and negative damping at high wind speeds. It confirms the suggestion that the amplification of frequency response at the natural frequency of the first tower modes is due to a linear instability with negative damping of this lateral bending mode. The damping is only slightly negative which is compensated by the nonlinear structural and aerodynamic forces, even for small lateral tower vibrations, causing the resulting loads to stay within the design range.

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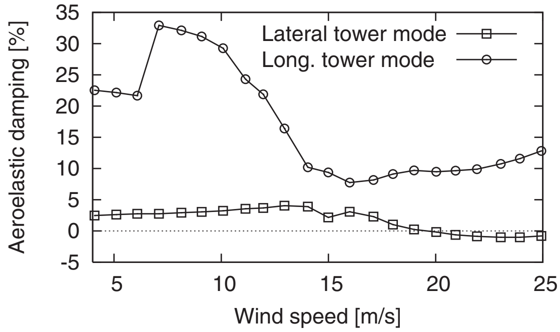

Figure 12. Aeroelastic damping of the first lateral and longitudinal tower bending modes as functions of wind speed computed with HAWCStab[11]

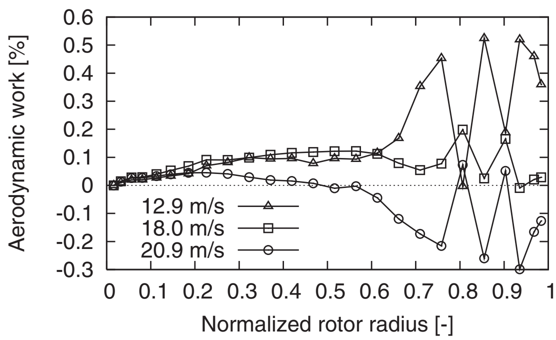

Figure 13. Aerodynamic work in the sections of a blade vibrating in the first lateral tower mode for different wind speeds computed with HAWCStab[11]

Figure 13 shows the work done by the linear aerodynamic forces on the sections of a blade integrated over one period of oscillation when the turbine is vibrating in the first lateral tower mode for three different wind speeds (the sign of the work is changed so that positive work corresponds to positive damping, i.e. the aerodynamic forces remove energy from the blade sections). The discontinuities at the 80 and 90% radii are caused by discontinuities in the aerodynamic characteristics of the NACA 63-415 airfoil due to stall-strips mounted on the leading edge of the blades around these radial sections to smoothen the stall characteristics (cf. Figure 8).

The distributions of the sectional aerodynamic work in Figure 13 show several features of the stall-induced tower vibrations: the main contribution to the work originates from the outboard part of the blade where the aerodynamic forces are largest. The negative damping of the lateral tower mode at $21\,\mathrm{m\,s^{-1}}$ is caused by aerodynamic work on the outboard part of the blade where the NACA 63-415 airfoil is used. The stall-strips around the 80 and 90% radii reduce this aerodynamic work, whereby the positive damping of the lateral tower mode around $13\,\mathrm{m\,s^{-1}}$ is reduced, but they also fulfill their purpose by increasing the negative damping at high wind speeds.

The solutions to the eigenvalue problem for the investigated turbine obtained by HAWCStab also contain the eigenvectors from which the mode shapes of the aeroelastic turbine modes are identified. Animations of the negative damped lateral tower mode at $22\,\mathrm{m\,s^{-1}}$ depicted in Figure 14 show that the tower top is vibrating in an almost 45 degree angle to the wind direction. Animation of the blade vibrations in this mode shape seen in a rotating view following the blade from the rotor plane shows that the direction of vibration is about –45 degrees relative to the rotor plane (with the sign definition as in Figure 5). This effective direction of vibration of the blade corresponds to the critical direction of vibration of the NACA 63-415 airfoil in stall, as shown in Figure 7 for an angle of attack of 20°.

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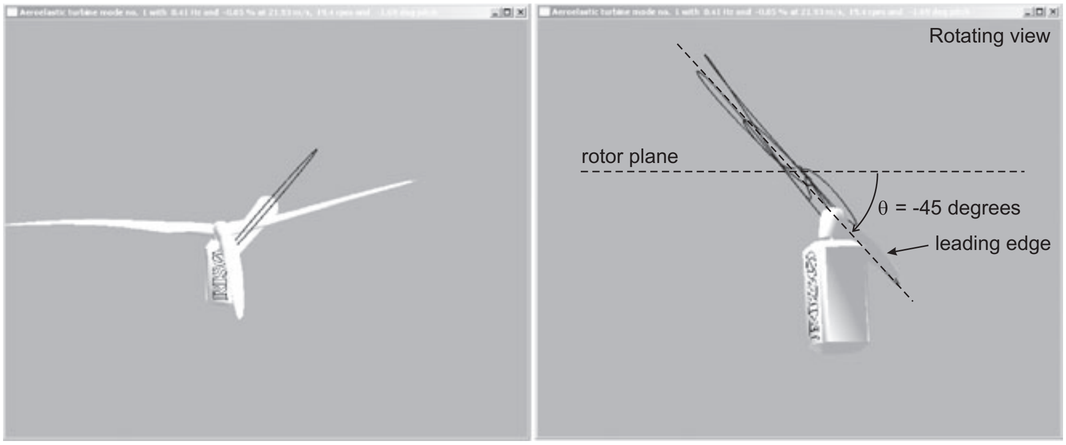

Figure 14. Snapshots of the negatively damped lateral tower mode at 22 m s^{-1}. Left picture is a top view of turbine with trace plotted for the tower top motion. Right picture is a rotating view of one blade, where the trace of tip motion shows the direction of blade vibrations due to the tower vibrations. Animations from HAWCStab[11]

The presented case study is a textbook example of stall-induced vibrations, and similar active-stall-regulated turbines have also experienced similar problems with tower vibrations. It is noted that the aerodynamic damping of the critical tower mode is not highly negative. Often problems can be solved by adding damping using a mechanical tower damper in combination with stall-strips.

Stall-Induced Edgewise Blade Vibrations

Another issue of stall-induced vibrations is the interaction of the blades with the remaining wind turbine as described in a previous work,[13] whereby the direction of blade vibrations in the turbine modes becomes highly dependent on the overall dynamics of the turbine.

Thomsen *et al*.[8] developed an experimental method for estimating the total damping of turbine modes of a turbine during operation. Figure 15 shows their damping estimates for the first edgewise whirling modes of the same stall-regulated 600 kW turbine used in the introduction of modal turbine dynamics. The experiment shows that the FW mode is more damped than the BW mode. The structural damping of these two modes is assumed to be the same because their natural frequencies and structural mode shapes are almost identical; the difference in total damping is therefore a difference in aerodynamic damping.

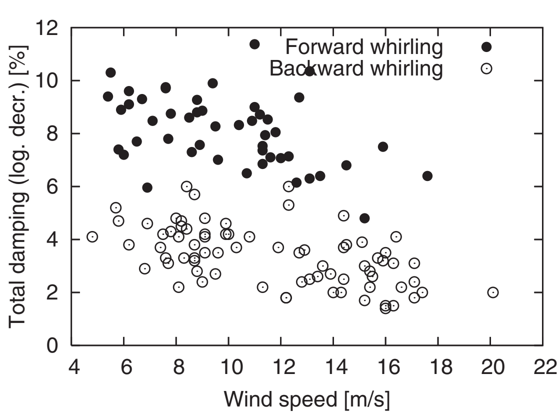

Figure 15. Aeroelastic damping of the first edgewise whirling modes of the 600 kW stall-regulated turbine measured in the experiment by Thomsen et al.[8]

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Aeroelastic stability analyses of this 600 kW turbine[11,14] agree with the experimental results that these aeroelastic turbine modes are differently damped by the aerodynamic forces. Figure 16 shows the measured and predicted aeroelastic damping of the first edgewise whirling mode pair. There are agreements in the level of damping, in the trend for increasing wind speed, and in the difference in damping of the FW and BW modes. The reason for the distinct damping of these two turbine modes that involve the same edgewise blade mode is that the directions of blade vibration in their mode shapes are different. Figure 17 shows traces of the blade tip motion for vibrations in the edgewise FW and BW modes of the 600 kW turbine during operation at $15\,m\,s^{-1}$. It shows that the blades vibrate more out of the rotor plane in the FW mode than in the BW mode. Both traces of the blade vibrations are elliptic, and using quasi-steady aerodynamics similar to the derivations of Equation (7), it can be shown that more out of plane opening of the elliptic blade motion leads to higher aerodynamic damping.[13] Hence, the difference in the measured aeroelastic damping of the edgewise whirling modes can be explained by the more out of the rotor plane vibrations of the blades in the FW mode.

This out-of-rotor-plane motion of the blades in the first edgewise FW mode is caused by the coupling to the second flapwise BW mode. As mentioned in the end of the introduction to modal turbine dynamics, the natural frequencies of these two whirling modes are so close at the operational rotor speed of 27 rpm (cf. Figure 2) that there is a coupling of their mode shapes. Figure 18 shows the FW and BW modal components (A_{BW} and A_{FW}, cf. Equation (2)) of the flapwise and edgewise blade deflections in structural modes number 7–10 of the 600 kW turbine as function of rotor speed. The whirling components for modes 7 and 10 show that these modes

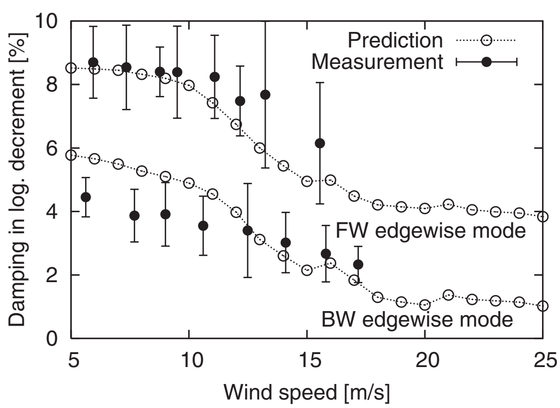

Figure 16. Measured and predicted damping of first edgewise whirling modes of the 600 kW stall-regulated turbine (taken from a previous study[11])

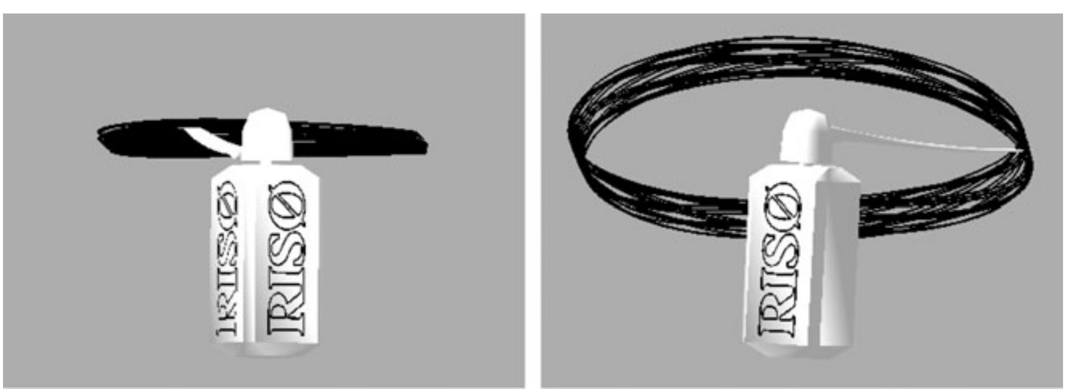

Figure 17. Traces of blade tip motion for vibrations in the edgewise BW (left) and FW (right) modes of the 600 kW turbine at $15\,m\,s^{-1}$. Animations from HAWCStab[11]

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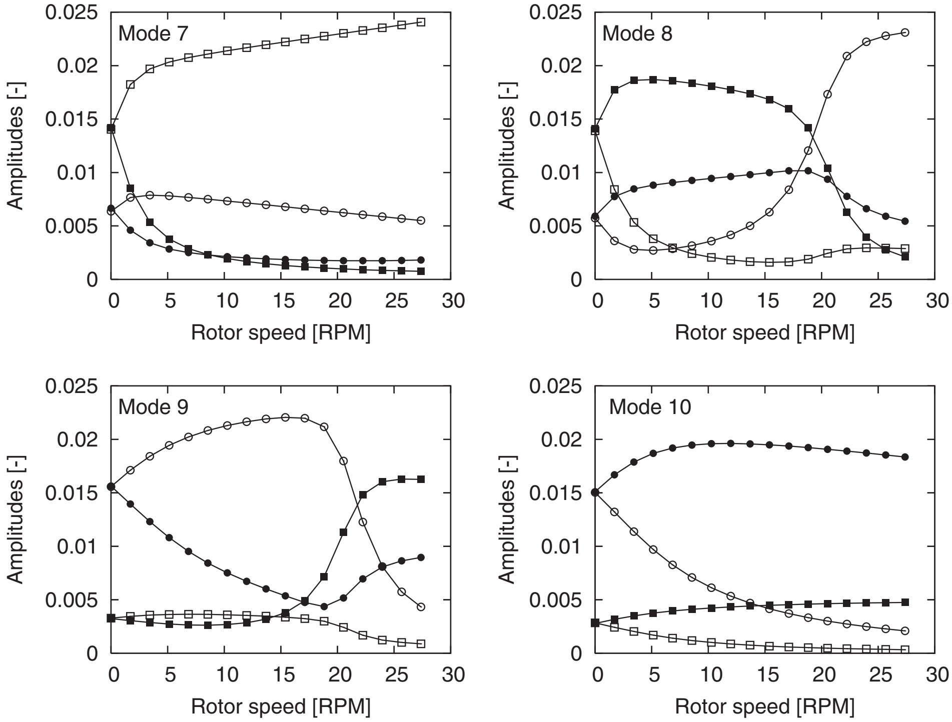

Figure 18. Modal whirling amplitudes for modes 7–10 of the 600 kW turbine, where the modes are sorted after frequency at each rotor speed (cf. natural frequencies in Figure 2). Flapwise whirling components are denoted A_{BW} = ○ and A_{FW} = ●. Edgewise whirling components are denoted A_{BW} = □ and A_{FW} = ■ (reproduced from a previous work[14])

are the first edgewise BW mode and the second flapwise FW mode, respectively. Modes 8 and 9 interchange mode shapes at the crossing of their natural frequencies around 20 rpm, whereby mode 9 at the operational speed of 27 rpm is the first edgewise FW mode. However, it has a large flapwise component from the modal coupling to the second flapwise BW mode, which explains the out-of-rotor-plane blade vibrations in this mode seen in Figure 17.

Hansen[13] showed that the coupling of the second flapwise and first edgewise whirling modes can be exploited to change the direction of blade vibration for both edgewise whirling modes away from the rotor plane. By for example increasing the edgewise bending stiffness of the blade, the natural frequencies of the first edgewise whirling modes may stay close to the natural frequencies of the second flapwise whirling, which increases the out-of-plane vibrations and thereby the aerodynamic damping of both edgewise whirling modes.

State-of-the-Art Issues

Although the trend for commercial wind turbines points towards pitch-regulated turbines, the risk of stall-induced vibrations is still an issue in the current research of turbine aeroelasticity. Here is a list of some of the topics that need to be investigated:

- *Stall-induced vibrations at standstill*. According to the IEC standards, the manufacturers must show that the parked turbine can withstand a 50 year wind coming from any direction. The most critical wind direction and resulting load may be determined by the risk of stall-induced vibrations. To assess the risk of stall-induced vibrations on parked turbines, it is necessary to consider cases where the flow over the airfoils is fully separated (deep stall). The largest uncertainty in this assessment is the airfoil characteristics in deep stall.

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- *Dynamic stall effects in deep stall*. Even when the steady-state airfoil characteristics have been identified from measurements, models of the unsteady aerodynamic forces in deep stall are still needed; current dynamic stall models reduce to quasi-steady aerodynamics for fully separated flows.
- *Structural damping enhancement*. The often light negative damping of turbine modes due to stall-induced vibrations can be compensated by structural damping. In the DAMPBLADE project[27] supported by the EC, it was shown that the material damping of the composite blade can be enhanced by careful design of the layups and material choices.
- *Active damping using trailing-edge flaps*. Active suppression of stall-induced vibrations using trailing edge flaps will increase the aerodynamic damping of low damped turbine modes and thereby reduce the fatigue loads. Preliminary design and application studies of trailing edge flaps for turbine blades are presented in Buhl *et al*.[28]

This list of research topics related to stall-induced vibrations is probably not complete. Hopefully, some readers may feel inspired to answer some of the open questions, or ask new questions based on this introduction to stall-induced vibrations.

Classical Flutter

Classical flutter is a much more violent instability than stall-induced vibrations. Here, the first torsional blade mode couples to a flapwise bending mode in a flutter mode through the aerodynamic forces: the change of angles of attack due to torsion changes the lift in an unfavorable phase with flapwise bending (see Figure 19). The flutter mode has highly negative damping which cannot be compensated by structural damping as the slightly negative damping of stall-induced vibrations.

There is yet no reports of actual flutter on wind turbines; however, the extensive studies of aircraft wing flutter (see e.g. textbooks on aeroelasticity[29,30]) have led to the question if flutter also may become a problem for wind turbine blades as they become more flexible. From those studies, it is assumed that a wind turbine may have the risk of flutter if the following main criteria are satisfied:

1. *Attached flow*. The flow over the blade must be attached to ensure that nose-up (towards stall) blade torsion leads to increased lift.
2. *High tip speeds*. The relative speed of the attached flow must be sufficiently high to ensure sufficient energy in the aerodynamic forces.
3. *Low stiffness*. The natural frequencies of a torsional mode and a flapwise bending mode must be sufficiently low for them to couple in a flutter mode.
4. *Aft center of gravity*. The center of mass in the cross sections on the outboard part of the blade must lie aft of the aerodynamic center to ensure the right phasing of the flapwise and torsional components of the flutter.

Other parameters such as the air–blade mass ratio, blade aspect ratio, material damping, and structural bending-torsion couplings (elastic and shear center positions) also influence the flutter limit; however, the listed criteria are the fundamental ones.

The first criterion shows that flutter may only become a problem for pitch-regulated turbines for which the blades are operating below stall. The second criterion shows that flutter may become a problem if the tip speeds

Figure 19. Example of a blade flutter mode illustrated by snapshots over one period of oscillation (taken from Hansen and Buhl[45])

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of pitch-regulated, variable-speed turbines are increased. Although, the tip speed is limited by noise and load requirements, it is still interesting to investigate the case when a turbine is experiencing an overspeed, or highly yawed inflow.

Lobitz[15] has investigated the flutter limit of a MW-sized wind turbine blade based on isolated blade stability analysis using quasi-steady and unsteady (Theodorsen) aerodynamics. He showed that the predicted flutter speed of the blade using quasi-steady aerodynamics is lower than the flutter speed obtained using unsteady aerodynamics. This increased flutter speed of the latter is caused by the decreased effective gradient of the lift curve (cf. the first criterion) when the induced velocities from the shed vorticity are included in the modeling of the unsteady aerodynamic forces.

Lobitz[16] has furthermore investigated the effects of moving the center of mass and reducing the torsional blade stiffness on the flutter limit of MW-sized blades using isolated blade analysis. He showed that the flutter mode for larger blades consists of the second flapwise and first torsional blade modes, and that the flutter speed limit decreases when the natural frequency ratio between these two modes is reduced (cf. the third criterion). Similarly, the flutter speed limit decreases when the center of mass is moved towards the trailing edge of the blade (cf. the fourth criterion).

Hansen[17] has also investigated the effect of reduced torsional blade stiffness on the risk of flutter for a MW-sized turbine using both isolated blade and full turbine analyses. He also found that the flutter mode consists of the second flapwise and first torsional blade modes, and that the aeroelastic damping of this flutter mode decreases when the natural frequency ratio between these two modes is reduced. The results from the isolated blade and full turbine stability analyses for the particular turbine are similar, suggesting that isolated blade analysis is sufficient for predicting the flutter limits. This simplification is convenient because in isolated blade analysis, one does not have to solve a system with periodic coefficients.

The following sections contain a description of the instability mechanism of classical flutter using a typical section model, and a case study of blade flutter on a three-bladed pitch-regulated turbine. The stability analysis of the typical section model gives a physical understanding of the flutter instability mechanism and leads to an expression for the flutter limit which confirms the main criteria for flutter listed above. The section ends with a list of the current issues on flutter.

Instability Mechanism for a Typical Section

To understand the instability mechanism of classical flutter, it is sufficient to consider a typical blade section with two degrees of freedom (see Figure 20) subjected to quasi-steady aerodynamic lift without *apparent mass* terms (see e.g. Fung[30]). The inflow to the airfoil is presumed parallel to the chord. The flapwise translation of the airfoil $h(t)$ is presumed perpendicular to the inflow and the torsional rotation $\theta(t)$ about a chord point in

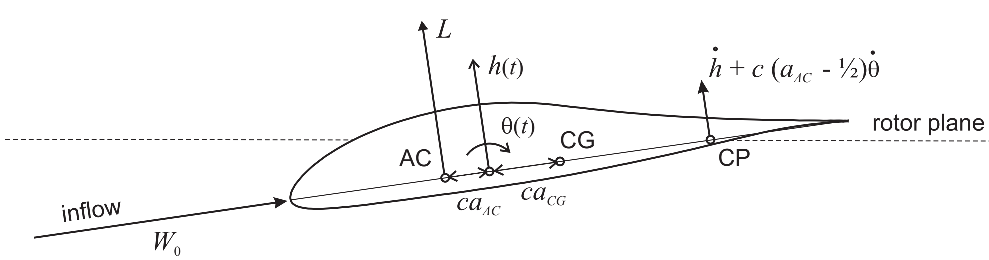

Figure 20. A typical blade section with two degrees of freedom: Flapwise translation h(t) perpendicular to the inflow and rotation $\theta(t)$ about a torsional point ca_{CG} in front of the center of gravity (CG) on the chord. The section is subjected to aerodynamic lift L at the aerodynamic center (AC) ca_{AC} in front of the torsional point. Angle of attack is defined at the collocation point (CP) in the three-quarter chord

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the distance ca_{CG} in front of the center of gravity (CG) on the chord. The section is subjected to aerodynamic lift L at the aerodynamic center (AC) in the distance ca_{AC} in front of the torsional point. The linear equations of motion can be derived as

$$\begin{aligned} &m\ddot{h} - mca_{CG}\ddot{\theta} + k_f h = L \\ &-mca_{CG}\ddot{h} + mc^2\left(r_{CG}^2 + a_{CG}^2\right)\ddot{\theta} + k_t\theta = ca_{AC}L \end{aligned} \tag{8}$$

where $(\dot{\cdot}) = d/dt$ denotes the time derivative, m is the mass per unit-length of the section, r_{CG} is the radius of gyration about CG normalized with the chord length c, and k_f and k_t are the flapwise and torsional stiffnesses, respectively.

Neglecting apparent mass terms, the quasi-steady aerodynamic lift L per unit-length is

$$L = \frac{1}{2}\rho c W^2 C_L(\alpha) \tag{9}$$

where ρ is the air density, W is the relative speed, α is the angle of attack, and C_L is the lift coefficient evaluated at α. To capture the effects of torsional velocity $\dot{\theta} \neq 0$ on the angle of attack, it is computed at the collocation point (CP) in the three-quarter chord.[31] The relative speed and the angle of attack are therefore (noting that the inflow is presumed parallel to the chord):

$$W = \sqrt{W_0^2 + \dot{h}^2} \quad \text{and} \quad \alpha = \arctan\left(\frac{W_0 \sin\theta - \dot{h} - c\left(\frac{1}{2} - a_{AC}\right)\dot{\theta}}{W_0\cos\theta}\right) \tag{10}$$

where W_0 is the steady-state relative speed of the inflow. Substitution into Equation (9), and linearization about $\theta = \dot{h} = \dot{\theta} = 0$ leads to the linear approximation to the lift:

$$L \approx L_0 + \frac{1}{2}c\rho W_0^2 C_L'\left[\theta - \frac{\dot{h}}{W_0} - \left(\frac{1}{2} - a_{AC}\right)\frac{c\dot{\theta}}{W_0}\right] \tag{11}$$

where the lift coefficient and its derivative $C_L' = dC_L/d\alpha$ are evaluated at $\alpha_0 = 0$, which for thin airfoils is $C_L' = 2\pi$. The steady-state lift L_0 may change the steady-state equilibrium of the section, but it has no influence on the stability of this equilibrium.

Neglecting the camber of the airfoil, the steady-state lift is zero ($L_0 = 0$), and the linear equations of motion (Equation (8)) with the linear lift (Equation (11)) can be written as

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = 0 \tag{12}$$

where the vector $\mathbf{x} = \{h/c, \theta\}^T$ contains nondimensional degrees of freedom. The structural mass, aerodynamic damping and aeroelastic stiffness matrices are

$$\mathbf{M} = \begin{bmatrix} 1 & -a_{CG} \\ -a_{CG} & r_{CG}^2 + a_{CG}^2 \end{bmatrix}, \quad \mathbf{C} = \frac{c\kappa}{W_0}\begin{bmatrix} 1 & \frac{1}{2} - a_{AC} \\ a_{AC} & a_{AC}\left(\frac{1}{2} - a_{AC}\right) \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} \omega_f^2 & -\kappa \\ 0 & r_{CG}^2\omega_t^2 - \kappa a_{AC} \end{bmatrix} \tag{13}$$

where $\omega_f = \sqrt{k_f/m}$ and $\omega_t = \sqrt{k_t/(mc^2 r_{CG}^2)}$ are the natural frequencies of the flapwise and torsional modes without inertia coupling ($a_{CG} = 0$), and $\kappa = \frac{\rho}{2m}W_0^2 C_L'$ is *aerodynamic stiffness* dependent of the air–section mass ratio ρ/m, the relative speed W_0, and the lift gradient C_L'.

For high relative inflow speeds W_0 and moderate frequencies of section vibrations ω as for wind turbine blades, the *reduced frequency* is small $k = c\omega/(2W_0) << 1$. Hence, the elements of the aerodynamic damping

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matrix **C** is an order smaller than the aerodynamic stiffness elements of the aeroelastic stiffness matrix **K** describing the lift change due to rotation of the section given by the torsion θ. Furthermore, the aerodynamic damping matrix is positive semi-definite $\mathbf{C} \geq 0$ if the aerodynamic center lies within the airfoil (more specifically if $0{\cdot}25 - \sqrt{17/4} < a_{AC} < 0{\cdot}25 + \sqrt{17/4}$). These small, purely dissipative aerodynamic forces are similar to material damping forces and are important for accurate predictions of flutter limits, but they are assumed not to have qualitative influence on the mechanism of the flutter instability, and they are therefore neglected in the following qualitative analysis of the flutter mechanism. Dissipative forces may have a destabilizing effect on systems other than non-gyroscopic, conservative systems,[32] but such an effect is assumed to be quantitative for the present *circulatory* system.

Neglecting the aerodynamic damping matrix $\mathbf{C} = 0$ and inserting the solution $\mathbf{x} = \mathbf{v}e^{\lambda t}$ into Equation (12) leads to the eigenvalue problem:

$$(\lambda^2\mathbf{M} + \mathbf{K})\mathbf{v} = \mathbf{0} \tag{14}$$

Non-trivial solutions ($\mathbf{v} \neq \mathbf{0}$) of this problem requires that it is singular, i.e. the determinant of $\boldsymbol{\lambda}^2\mathbf{M} + \mathbf{K}$ is zero leading to the characteristic equation:

$$r_{CG}^2\lambda^4 + \left[\left(r_{CG}^2 + a_{CG}^2\right)\omega_f^2 + r_{CG}^2\omega_t^2 - \kappa\left(a_{AC} + a_{CG}\right)\right]\lambda^2 + \omega_f^2\left(r_{CG}^2\omega_t^2 - \kappa a_{AC}\right) = 0. \tag{15}$$

Its zeroes are the eigenvalues of equation (14), which generally are complex $\lambda = \beta + i\omega$. If the real part of one eigenvalue is positive $\beta > 0$, then the equilibrium of the section is unstable, because the solution $\mathbf{x} = \mathbf{v}e^{\lambda t} = \mathbf{v}e^{\beta t}(\cos\omega t + i\sin\omega t)$ in this case will grow exponentially in time. Stability limits for the section are therefore defined by the parameters where the real part of an eigenvalue becomes positive. The Routh–Hurwitz criteria[32,33] state that the real part of all zeroes of the polynomial in Equation (15) are negative if its coefficients are positive:

$$\left(r_{CG}^2 + a_{CG}^2\right)\omega_f^2 + r_{CG}^2\omega_t^2 - \kappa\left(a_{AC} + a_{CG}\right) > 0 \quad \text{and} \quad r_{CG}^2\omega_t^2 - \kappa a_{AC} > 0. \tag{16}$$

On the limit of the first criterion, there is one complex eigenvalue with a positive real part, because the characteristic Equation (15) corresponds $\lambda^4 = -\gamma^2$ (γ is a real constant) if the second criterion is satisfied. The non-zero imaginary part of this complex eigenvalue shows that the instability is oscillatory, hence this criterion defines the flutter limit of the section as

$$\frac{\rho}{2m}W_0^2 C_L' < \omega_f^2\frac{r_{CG}^2 + a_{CG}^2}{a_{AC} + a_{CG}} + \omega_t^2\frac{r_{CG}^2}{a_{AC} + a_{CG}} \quad \text{for} \quad a_{AC} + a_{CG} \geq 0 \tag{17}$$

where the aerodynamic stiffness κ has been inserted. This inequality must be turned for $a_{AC} + a_{CG} < 0$. The second criterion in Equation (16) defines the divergence limit of the section as

$$\frac{1}{2}c\rho W_0^2 C_L' c a_{AC} < k_t \tag{18}$$

where κ and ω_t have been inserted. On this limit, the structural and aerodynamic torsional stiffnesses (lower right element of **K**) cancel out. Beyond this limit, an increase in torsion will increase the lift which again increases the torsion, leading to divergence.

The simple analytical expression of the flutter limit in Equation (17) derived from the simple typical section model confirms the main criteria for the risk of flutter listed in the introduction. The flutter may occur under attached flow conditions $C_L' > 0$ if the air–blade mass ratio represented by ρ/m and the tip speed represented by W_0 are sufficiently high for the aerodynamic forces to overcome the dynamic elastic forces represented by the uncoupled flapwise and torsional natural frequencies ω_f and ω_t. These uncoupled frequencies are weighted by the coupling factors $(r_{CG}^2 + a_{CG}^2)/(a_{AC} + a_{CG})$ and $r_{CG}^2/(a_{AC} + a_{CG})$ that may go to infinity if the center of gravity lies in the aerodynamic center where $a_{AC} + a_{CG} = 0$. Furthermore, if the center of gravity lies in front of the aerodynamic center ($a_{AC} + a_{CG} < 0$) then the turned inequality will always be satisfied for attached flow $C_L' > 0$, which is why mass added to leading edges of aircraft wings is a practical solution to flutter problems.

Figure 21 shows in the left plot a stability diagram with flutter (and divergence) speed limits for a typical section of a wind turbine blade as functions of the chordwise position of the center of gravity for different torsional frequencies 6, 8 and 10 Hz. Apparent mass terms and unsteady aerodynamics are included in the computation of these stability limits using a state-space approximation of the Theodorsen theory.[34]

The dashed curve labeled '6 Hz-QS' is the quasi-steady approximation of the flutter limit in Equation (17), which is lower than in the case of unsteady aerodynamics. This agrees with the conclusion by Lobitz[15] that quasi-steady aerodynamics reduces the flutter speed; although in the present case, some of this reduction is due to the neglecting of the aerodynamic damping terms in the derivation of Equation (17). The neglecting of the added mass terms will have a small stabilizing effect on the flutter speed because it increases the frequencies.

All flutter speed limits have a vertical asymptote when the center of gravity lies at the aerodynamic center. The flutter speeds reduce as the center of gravity is moved aft on the section which is directly related to the coupling factor $r_{CG}^2/(a_{AC} + a_{CG})$ on the uncoupled torsional frequency ω_t in Equation (17). The coupling factor on the uncoupled flapwise frequency ω_f has a minor effect because the torsional frequency is three times higher. Hence, the main reduction in flutter speed as the center of gravity is moved aft is due to the increased flapwise–torsional coupling. However, a part of the flutter speed reduction is due to the decreased torsional frequency as the moment of inertia about the torsional point (EA) increases for increasing distance to the center of gravity.

Figure 21 shows in the right plot the eigenvalues of the flapwise and torsional modes as function of relative speed for a section with an uncoupled torsional frequency of 6 Hz and the center of gravity at the 40% chord point. A flutter mode with negative damping arises around 62 m s^{-1} which agrees with the flutter limit in the left plot for this torsional stiffness and center of gravity. The frequencies of the flapwise and torsional modes approach each other as the relative speed is increased from zero and almost coincide at the flutter limit, which is a typical behavior for the frequencies of the two modes that couple in the flutter mode. For a lower initial uncoupled torsional frequency, this coincidence between the coupled flapwise and torsional frequencies occurs at a lower flutter speed (cf. left plot), because lower aerodynamic forces are needed for overcoming the elastic forces that resist the coupling of the flapwise and torsional modes in the flutter mode.

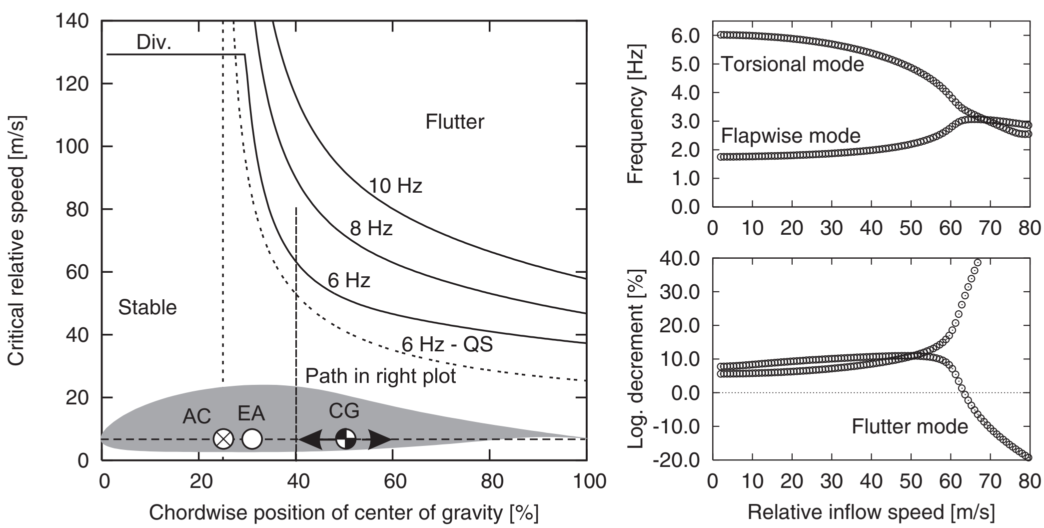

Figure 21. Left plot: flutter and divergence speed limits for a typical section of a wind turbine blade as functions of the chordwise position of the center of gravity for different torsional frequencies 6, 8 and 10 Hz. Right plot: eigenvalues of flapwise and torsional modes as function of relative speed for 6 Hz and center of gravity at 40% chord point. Apparent mass terms and unsteady aerodynamics using Theodorsen theory are included, except for dashed curve labeled '6 Hz-QS' which is based on Equation (17)

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Another characteristic of flutter is that the lowest damping of the modes suddenly becomes negative as the flutter mode arises; the critical speed range is small which in the early days of aviation has given pilots a fatal surprise. This narrow critical speed range also leads to the questions if a blade on a wind turbine running at nominal rotor speed but with a large yaw misalignment in high winds can experience flutter on the part of the azimuth rotation where its blades meet the incoming wind leading to a higher relative speed.

Case Study—Blade Flutter on a 5 MW Turbine

This section contains a short case study of blade flutter on the virtual 5 MW pitch-regulated turbine designed by NREL.[18] The turbine has a 126-m-diameter rotor with 61·5-m-long blades with a natural frequency of the first torsional mode around 8 Hz. The blade design is closely related to an actual blade with the same approximate distributions of bending and torsional stiffnesses, center of gravity, and mass per unit-length along the blade span.

Figure 22 shows the scaled planform of the 61·5 m blade with the chordwise positions of the aerodynamic center and the center of gravity. It shows that the center of gravity lies aft of the aerodynamic center, hence the blade will experience flutter if the relative speed of the inflow to the blade is sufficiently high.

The nominal rotor speed is 12·1 rpm. It is interesting to investigate how far this operational speed of the turbine is from the flutter speed. This investigation can be performed with aeroelastic stability tools for wind turbines.[11,12,15,35,36] These tools are based on aeroelastic eigenvalue analysis, where the nonlinear aeroelastic model of the turbine is linearized about a steady-state equilibrium to set up an eigenvalue problem. The method is similar to the eigenvalue problem for the typical section in equation (14) except that these tools include all degrees of freedom of the turbine and all terms of the unsteady aerodynamics.

The present flutter analysis is based on the stability tool HAWCStab.[11] To ensure attached flow condition along the blades, the operational conditions are set to no wind and zero blade pitch, whereby the angles of attack along the blade become low. To further ensure that the rotor speed is the only varying parameter in this analysis of determining the flutter speed, the linearization of the nonlinear aeroelastic model is performed about the undeformed blade (in reality, the blade would deform under the steady-state aerodynamic loads at high rotor speeds).

Figure 23 shows the natural frequencies and damping of the aeroelastic blade modes (left plots) and turbine modes (right plots) obtained from isolated blade and full turbine analyses, respectively, as function of rotor speed. Modes involving flapwise bending and torsional blade modes may couple in the flutter mode, whereas modes involving edgewise blade modes, tower bending modes and drivetrain torsion modes have no influence on the flutter mode, and they are therefore not plotted for the turbine modes.

In the isolated blade analysis, a flutter mode obtains a negative damping around 24 rpm. The flapwise bending and torsional modes that couple in this flutter mode are emphasized in the left plots by a larger line thickness. It is the third flapwise bending mode that couples with the first torsional mode; their natural frequencies almost coincide, similarly to the two modes of the typical section in Figure 21. The natural frequency of the first flap-

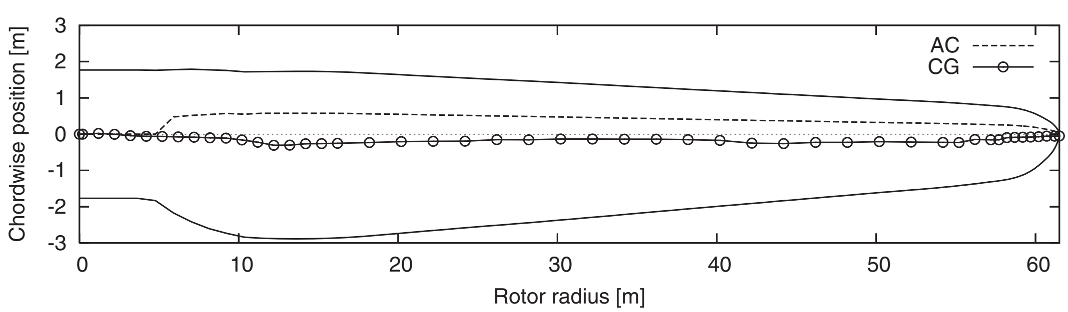

Figure 22. Scaled planform of the NREL 5 MW blade showing the positions of the aerodynamic center (AC) and the center of gravity (CG). Reproduced from Jonkman[18]

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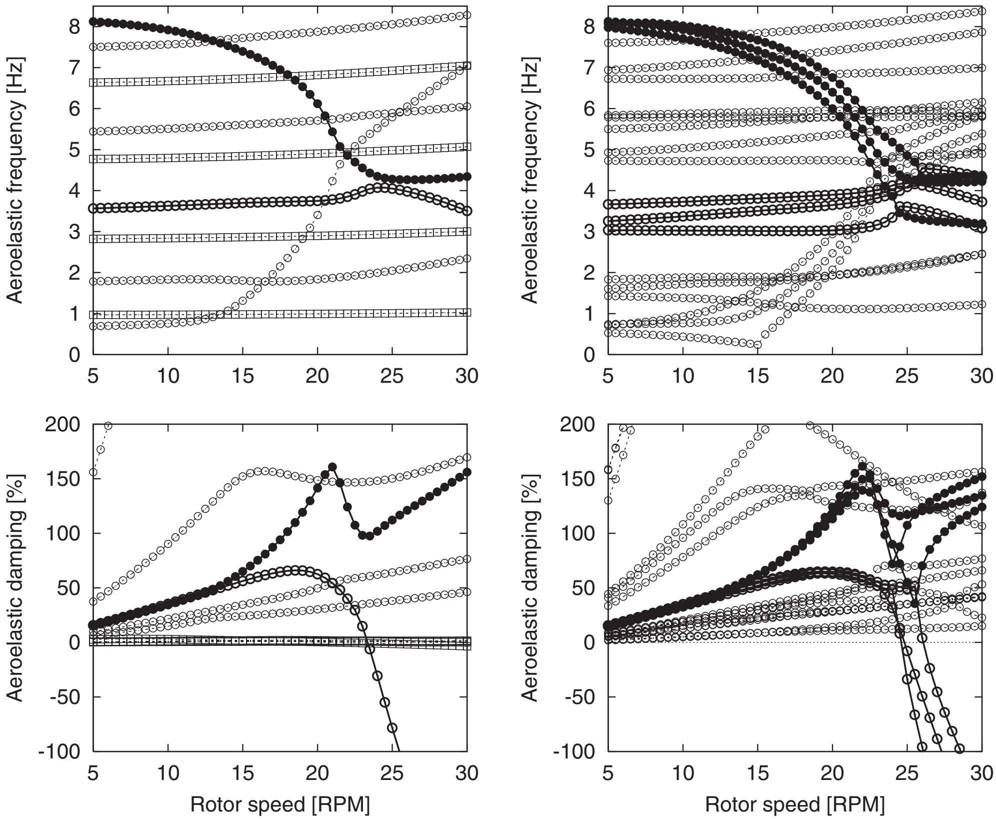

Figure 23. Natural frequencies (top plots) and damping (bottom plots) of the aeroelastic blade modes (left plots obtained from isolated blade analysis) and turbine modes (right plots obtained from full turbine analysis) as function of rotor speed. Operational conditions are no wind, zero blade pitch and undeformed steady state. Flapwise modes denoted by ○ and torsional modes denoted by ●. Edgewise blade modes denoted by □. Remaining turbine modes not plotted. Computations from HAWCStab[11]

wise bending mode is highly affected by the increasing rotor speed due to the aerodynamic coupling with blade torsion; however, it is not the flapwise mode that directly couples with the torsional mode in the flutter instability.

In the full turbine analysis, three flutter modes obtain negative damping at 25–27 rpm. There are clear similarities with the results of the isolated blade analysis: The three rotor modes (symmetric and two whirling modes, cf. the Introduction) involving the third flapwise blade mode and the three rotor modes involving the first torsional blade mode couple in the three flutter modes. These flapwise and torsional rotor modes couple in pairs, where the third flapwise symmetric rotor mode couples with the symmetric torsional rotor mode, the third flapwise BW rotor mode couples with the BW torsional rotor mode, and the third flapwise FW rotor mode couples with the FW torsional rotor mode. Similar to the first flapwise blade mode of the isolated blade, the natural frequencies of the first flapwise symmetric and whirling rotor modes increase with rotor speed due to the aerodynamic coupling with blade torsion, but they are not directly part of the flutter modes.

Figure 24 illustrates the similarity between the flutter instability of the isolated blade and full turbine by four snapshots of flutter mode over one period of oscillation for a rotor speed of 25 rpm. The aerodynamic modal forces due to the vibrations in the flutter modes are plotted to show the mechanism of the flutter instability: these aerodynamic forces act in the same direction as the blade motion and add energy into the vibrations. The forces in the turbine flutter mode are slightly smaller than in the blade flutter, corresponding to the more negative damping of the blade flutter mode (cf. Figure 23).

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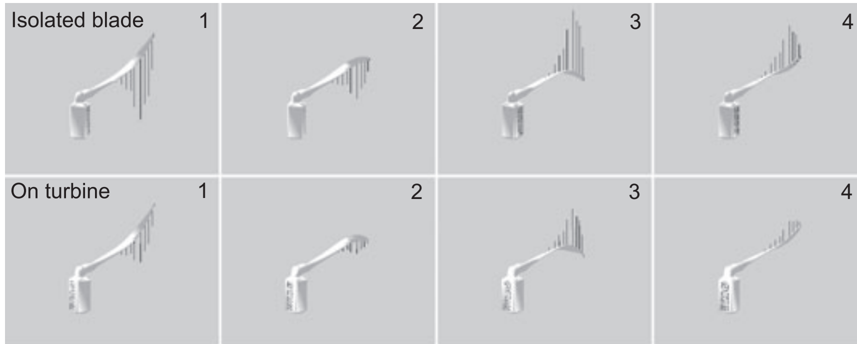

Figure 24. Blade flutter modes with aerodynamic modal forces obtained from isolated blade analysis (top) and full turbine analysis (bottom) at a rotor speed of 25 rpm (cf. Figure 23) illustrated by exaggerated vibrations over one period of oscillation. Animations from HAWCStab[11]

The presented case study has similarities with previous studies of wind turbine flutter, e.g. the flutter speed of the rotor is twice the nominal rotor speed as also found by Lobitz[15] for a 1·5 MW turbine, and the flutter limit predicted by isolated blade analysis is similar to the flutter limit predicted by full turbine analysis as also found by Hansen[17] for another MW-sized turbine. The present case study leaves several questions unanswered, partly because of limitation of the scope, but also because the study of wind turbine flutter is still a new field of research in wind energy; here follows a short and probably not complete list of the current issues in this field.

Many smaller turbines?

State-of-the-Art Issues

The increased risk of flutter due to the increased size and optimization of pitch-regulated wind turbines makes flutter an important issue in the current research of turbine aeroelasticity. Here is a list of some of the topics that need to be investigated:

- *Flutter experiments*. No experimental data are currently available for validation of the flutter limits predicted for conventional three-bladed turbines using the aeroelastic stability tools. Such experiments could be performed by controlling the generator torque to carefully letting a turbine overspeed until flutter occurs. To avoid the flutter to develop fully whereby the blades probably would fail, a so-called *flutterometer* known from aeronautics[37] could be used to detect the flutter before it occurs.
- *Near wake effects on flutter limits*. Unsteady aerodynamic effect of the shed vorticity due to angle of attack variations on the flutter limit is modeled by the Theodorsen theory in state-of-the-art stability analyses. A similar unsteady aerodynamic effect arises due to variations in the trailing vorticity of the near wake as a blade vibrates.[38] This near wake effect depends on the radial distribution of the lift, and it is largest close to the blade tip. This spanwise dependency of the unsteady aerodynamic forces is assumed to affect the flutter limits.
- *Flutter limits for yaw misaligned inflow*. Flutter is a violent instability with a narrow critical speed range where a highly negative damped flutter mode arises. It should be investigated if a wind turbine in large yaw misalignment with the wind may experience flutter on the part of the azimuth blade rotation where the blades meet the incoming wind leading to a higher relative speed. What is a sufficient separation between the nominal and flutter rotor speed to avoid this risk of partial flutter for large yaw errors and high winds?
- *Nonlinear blade coupling effects on flutter limits*. Modern slender blades are subjected to large flapwise deflections due to the aerodynamic thrust, whereby a nonlinear geometric coupling between edgewise bending and torsion arises due to the flapwise curvature.[39,40] This coupling will affect the flutter limits because the torsional component of the flutter mode will be combined with edgewise bending. Preliminary work at

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Risø National Laboratory with a blade model including the edgewise–torsional coupling[41] shows that the flutter speed decreases for increased flapwise deflection.[42]

- *Pitch–torsion coupling in flutter modes*. The torsional flexibility of the pitch system due to compressibility of hydraulic liquids, or free play in electromechanical pitch gear drives is assumed to affect the flutter limits. Preliminary work at Risø National Laboratory with a detailed model of a hydraulic pitch system shows that a new torsional rigid-body mode couples to the blade; however, further investigations including pitch bearing friction is needed to determine the effects of this mode on flutter.
- *Active damping using trailing-edge flaps*. Active suppression of flutter using trailing edge flaps is a research technology on aircraft wings, which is directly applicable for wind turbine blades if operational limitations due to flutter problems should arise here. Preliminary design and application studies of trailing edge flaps for turbine blades are presented in Buhl *et al*.[28]
- *False linear flutter limits due to large nonlinearities*. Large nonlinearities such as free play in the blade rotation at the pitch bearing may change the flutter instability from a supercritical to a subcritical Hopf bifurcation, i.e. there is no stable solution branch in continuation of the steady-state solution at the point when it losses its stability. Instead, there may be a stable limit-cycle solution with large amplitude that appears below this point of linear instability. Hence, a linear stability analysis will lead to a false prediction of the flutter limit because disturbances will push the blade into this limit-cycle flutter solution when it appears.[43,44]

This list is probably not complete, and there will arise further issues if flutter becomes a real problem for wind turbines. Hopefully, the present introduction to flutter of wind turbines can inspire some readers to investigate these issues before such problems occur.

Conclusion

The two types of aeroelastic instabilities for modern commercial wind turbines, *stall-induced vibrations* and *classical flutter*, are reviewed and explained based on previous works, derivations of analytical stability limits for typical blade sections, and case studies of different wind turbines. The fundamental mechanisms of the instabilities are illustrated, identifying the important parameters for assessing the risk of these instabilities.

Stall-induced vibrations may occur on a stall-regulated turbine, if the blade airfoils have abrupt stall characteristics with high maximum lift and low drag, if the blades vibrate in certain directions relative to the inflow, and if the structural damping is sufficiently low. Even though the commercial trend points towards pitch-regulated turbines, the risk of stall-vibrations is still an important issue. Turbines must withstand a 50 year wind coming from any direction in a parked situation, where for example the flow over a blade may be separated and in an unfavorable direction relative to the direction of vibration of a blade, which is therefore excited. Stall-induced vibrations at standstill are a current research issue because the airfoil characteristics and dynamic behavior in deep stall are unknown parameters needed for the stability analysis of such situation.

Flutter may occur on a pitch-regulated turbine, if the blade tip speed becomes sufficiently high due to an overspeed, or large yaw misalignment, if the torsional blade stiffness is sufficiently low, and if the center of gravity lies sufficiently aft in the blade cross sections along the blade span. The reported work on wind turbine flutter is still limited, and no experimental data for modern commercial turbines are available for validation of the predicted flutter limits. Current research issues are the effects of large flapwise blade deflections and the unsteady aerodynamic effects from the near wake of trailing vorticity on these predicted flutter limits.

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